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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 01/07/2016 |
| **PI Name:** | Kronenberg, Amy Sc.D. | | |
| **Project Title:** | Comparative Analysis of Charged Particle-Induced Autosomal Mutagenesis in Murine Tissue and Cells | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 94720-8099 | **Congressional District:** | 13 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2011 Space Radiobiology NNJ11ZSA001N |
| **Start Date:** | 03/14/2012 | **End Date:** | 03/31/2016 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 4 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Simonsen, Lisa | **Contact Phone:** | |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Extended to 3/31/2016 per S. Monk/LaRC (Ed., 9/14/15) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Turker, Mitchell ( Oregon Health & Science University )<br>Grygoryev, Dmytro ( Oregon Health & Science University )<br>Grossi, Gianfranco ( Universita Federico II )<br>Gauny, Stacey ( Lawrence Berkeley National Laboratory )<br>Ohlrich, Anna ( Oregon Health & Science University )<br>Hryciw, Gwen ( Oregon Health & Science University ) | | |
| **Grant/Contract No.:** | NNJ12HB88I | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Astronauts in space are exposed to a variety of charged particle radiations, some of which travel at very high velocities that can penetrate spacecraft, damage tissues, and increase the risk of cancer. This proposal addresses two cancer development processes, genomic instability and aneuploidy, links them to each other and to specific mechanisms of mutagenesis at an endogenous autosomal locus in epithelial cells. The aims proposed use a comparative biology approach and a transgenic mouse model that involves both in vitro and in situ approaches. This strategy enables quantitative, coupled analyses of early changes in cells that survive the exposure and a direct test of the relevance of these changes to the development of persistent changes in situ. The studies address the radiation quality dependence of genomic instability and aneuploidy focusing on charged particle beams spanning a critical linear energy transfer (LET) range from 25 keV/µm to 151 keV/µm, and will define the shape of the dose-response for these endpoints, particularly at low fluence. The rationale for working in this range of LET is the observation of increased biological effectiveness for cancer relevant endpoints in our previous studies, including mutagenesis in human and rodent cells, and work by others concerning Harderian gland tumorigenesis in mice. The investigations test the hypothesis that the induction of genomic instability and aneuploidy will increase as a function of ionization density for low fluence exposures, leading to genome-wide alterations in viable, mutant cells found in normal tissue. This project will define links between the processes that lead to early establishment of genomic instability, aneuploidy, and mechanisms of autosomal mutagenesis found in many kinds of human epithelial cancers. These quantitative, mechanism-based studies will aid in the extrapolation of results across species to help reduce the uncertainties in cancer risk projections for astronauts exposed to spaceflight radiations. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research examines the effects of charged particle radiations with different ionization densities using normal mammalian epithelium. The research is designed to help determine links between cancer development processes at relatively low doses. This work is of relevance to naturally occurring Earth-based exposures to densely ionizing radiations such as radon. It will also inform our understanding of the mechanisms underlying potential risks to normal tissues associated with the implementation of charged particle radiation therapy. |
| **Task Progress:** | This project was initiated in Spring 2012. During the third year of funding, we conducted additional experiments to determine the dose-response for autosomal mutagenesis following exposure to graded doses of 250 MeV/amu O ions (LET = 25 keV/µm) using cultured kidney cells from B6D2F1 mice. A linear dose-dependent increase in Aprt mutations was noted for B6D2F1 Aprt +/- cells exposed in Summer 2014 at the NASA Space Radiation Laboratory (NSRL) accelerator, similar to what was found for the initial O ion exposures conducted in 2013. Oxygen ions were less mutagenic per unit dose than Si ions (240 MeV/amu, LET = 78 keV/µm), but substantially more mutagenic than 1 GeV protons (LET=0.24 keV/µm).<br>Replicate flasks of B6D2F1 Aprt +/- kidney epithelial cells were exposed to 160 cGy of O ions and Aprt mutant collections were initiated for future molecular studies. A parallel set of non-mutant B6D2F1 kidney cell clones was also isolated following exposure to 160 cGy of O ions. In Fall 2014, replicate flasks of B6D2F1 Aprt +/- cells were exposed to either 80 cGy of O ions or 40 cGy of O ions. These cultures were archived for future studies on mutational spectra as a function of dose.<br><br>Studies begun in year 2 with B6D2F1 Aprt +/- kidney epithelial cells exposed in situ to 250 MeV/amu O ions were completed. The linear dose-response for O ion-induced mutations in B6D2F1 Aprt +/- cells exposed to O ions of the same LET was conserved if the kidney epithelial cells were exposed in situ and harvested 3-5 months later. The dose-response study for the induction of Aprt mutants in kidney cells exposed in situ was similar to the dose-response for Aprt mutation in cultured kidney cells exposed to O ions with a similar LET. Oxygen ions were less mutagenic to primary kidney epithelial cells than Si ions (LET = 78 keV/µm), which is in agreement with the results from studies using cultured kidney epithelial cells. Primary kidney clones were collected at all dose levels (0, 40, 80, 160 cGy) to determine the spectrum of mutations induced by graded doses of O ions.<br><br>During this funding period we also continued our analysis of the molecular bases for the formation of Si ion-induced Aprt kidney cell mutants from primary kidney clones (LET=78 keV/µm). We completed the analysis of changes along chromosome 8 for Aprt mutants isolated after exposure to 0.8 Gy (53 mutants) and after exposure to 0.4 Gy (65 mutants). These results are being compared with the results from a similar group of mutants isolated after exposure to 1.6 Gy of the same Si beam and to the results from a large sample of spontaneous Aprt mutants. |
| **Bibliography Type:** | Description: (Last Updated: 11/01/2023) |
| **Abstracts for Journals and Proceedings** | Grygoryev D, Dan C, Gauny S, Ecklemann B, Ohlrich A, Connolly M, Lasarev M, Grossi G, Turker MS, Kronenberg A. "Autosomal mutants of proton-exposed kidney cells display loss of heterozygosity on other chromosomes." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014.<br>2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014.<br>http://www.hou.usra.edu/meetings/hrp2014/pdf/3214.pdf , Feb-2014 |
| **Abstracts for Journals and Proceedings** | Hryciw G, Grygoryev D, Ohlrich A, Dan C, Lasarev M, Gauny S, Kronenberg A, Turker M. "Induction of radiation signature mutations in vivo with charged particles at low fluence." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014.<br>2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014.<br>http://www.hou.usra.edu/meetings/hrp2014/pdf/3025.pdf , Feb-2014 |
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| **Abstracts for Journals and Proceedings** | Kronenberg A. "High LET-induced mutagenesis." 13th International Workshop on Radiation Damage to DNA, Cambridge, MA, June 14-18, 2014.<br>13th International Workshop on Radiation Damage to DNA, Cambridge, MA, June 14-18, 2014. , Jun-2014 |

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| **Abstracts for Journals and Proceedings** | Kronenberg A, Gauny S, Grossi G, Grygoryev D, Ohlrich A, Turker MS. "Linking chromosome aberrations and mutations in epithelium exposed to charged particle radiation of importance in space flight." 11th International Symposium on Chromosome Aberrations (ISCA11), Rhodes, Greece, September 12-14, 2014.<br>Abstract book. 11th International Symposium on Chromosome Aberrations (ISCA11), Rhodes, Greece, September 12-14, 2014. p. 17. , Sep-2014 |
| **Abstracts for Journals and Proceedings** | Kronenberg A, Gauny, S, Grossi G, Grygoryev D, Ohlrich A, Turker MS. "Comparative analysis of charged particle-induced autosomal mutations." European Radiation Research Society, Rhodes, Greece, September 2014.<br>European Radiation Research Society, Rhodes, Greece, September 2014. , Sep-2014 |
| **Abstracts for Journals and Proceedings** | Kronenberg A, Gauny S, Grossi G, Grygoryev D, Ohlrich A, Turker MS. "Widespread genomic change after exposure of epithelium to charged particles." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014.<br>60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014 |
| **Abstracts for Journals and Proceedings** | Kronenberg A, Gauny S, Grossi G, Dan C, Grygoryev D, Turker M. "Genotoxicity of charged particles of importance in space flight using murine kidney epithelial cells." Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013.<br>Journal of Radiation Research. 2014 Mar;55(Suppl 1):i77–i78. (Proceedings of Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013.) http://dx.doi.org/10.1093/jrr/rrt191 ; PubMed Central PMCID: PMC3941538 , Mar-2014 |
| **Articles in Peer-reviewed Journals** | Grygoryev D, Dan C, Gauny S, Eckelmann B, Ohlrich AP, Connolly M, Lasarev M, Grossi G, Kronenberg A, Turker MS. "Autosomal mutants of proton-exposed kidney cells display frequent loss of heterozygosity on nonselected chromosomes." Radiat Res. 2014 May;181(5):452-63. http://dx.doi.org/10.1667/RR13654.1 ; PubMed PMID: 24758577 , May-2014 |
| **Books/Book Chapters** | Goodhead D, Preston RJ, Fleming PA, Held KD, Kronenberg A, Nelson GA, Schimmerling W, Shaw RP, Weil MM. "Commentary No. 23. Radiation protection for space activities: Supplement to previous recommendations." Bethesda, MD : National Council on Radiation Protection and Measurements, 2014. ISBN 978-0-9835450-6-4S. See http://www.ncrppublications.org/index.cfm?fm=Product.AddToCart&pid=6775013616 for ordering., Nov-2014 |